Payload-Aware Open-Set Interference Recognition for GEO Bent-Pipe Transponders

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Abstract

Ground monitoring of a transparent geostationary satellite observes interference after propagation coupling, transponder filtering, nonlinear amplification and receiver imperfections. Consequently, waveform labels need not correspond to uniquely identifiable spectral causes. We formulate a controlled, payload-aware open-set recognition study that distinguishes three sources of monitoring information: dual-polarization observations, an observation-derived cancellation residual, and operator carrier-plan masks. A reference-model simulator produces paired multicarrier scenes with eight known classes, three near-family unknowns, an equivalence-limited noise family and a benign carrier-identification control. Independent scenario roles separate weight fitting, checkpoint selection, rejection calibration and evaluation. Three canonical views and a separately matched plan-mask comparison define a required fifteen-fit campaign. Unknown-positive rankings are distinguished from known-only calibrated operating points; uncertainty resamples whole scenario groups while retaining the trained models and thresholds. Existing simulated visibility and cancellation diagnostics motivate the experiment but do not establish recognition superiority. Production-scale recognition measurements are NOT RUN, the primary scorer is not yet researcher-designated, and prior exposure of a small part of the preserved test population is disclosed. This draft supplies the complete physical and statistical methods, auditable existing evidence, and publication-ready result specifications without presenting unexecuted comparisons as findings.

Manuscript status. This is the original GEO-focused experimental manuscript, separate from the repository's measured-GNSS/comparative-methods paper and its independent literature review. The physical and statistical methods below are implemented or explicitly identified as specifications. The required production-scale recognition campaign and separately authorized final evaluation are **NOT RUN**. Comparisons are exploratory under the accepted statistical design: the word *hypothesis* identifies a scientific question, not a completed confirmatory superiority test. Evidence identifiers E01–E18 resolve in `geo_b2_evidence.md`; its companion JSON binds the inspected source files and reused figures by SHA-256. This manuscript baseline is main `6e9ff7f1c29a33e813015753cc311080da2a77ff`. No protected observation was opened to write this draft.

1. Introduction

Ground-based interference monitoring of a geostationary (GEO) bent-pipe link is an inverse problem with an important observation boundary. An interfering uplink can traverse the wanted satellite's input multiplexer, amplifier and output multiplexer, whereas an interfering signal added on the downlink can bypass that payload. Their received spectra may nevertheless resemble one another. Transparent-channel reference models make the ordering of these operations explicit; they do not, by themselves, provide a unique inverse mapping from a power spectrum to an interference source (ETSI 2014, 2021). The present study therefore treats source recognition as a conditional statistical task under stated observation assumptions, rather than as identification of an orbital origin from a spectral image. [E01, E02]

Existing satellite monitoring research includes weak-interference detection after signal cancellation, prediction of normal received spectra, and learning from real satellite captures combined with terrestrial interferers (Politis et al. 2017; Pellaco, Singh, and Jaldén 2019; Henarejos, Vázquez, and Pérez-Neira 2019). Thus neither applying machine learning to satellite interference nor subtracting an estimated desired signal is a new contribution here. The question is more specific: which monitoring information supports recognition when weak external signals,

polarization leakage and nonlinear payload effects coexist, and when some evaluation families were absent from classifier training?

Operator knowledge creates a useful but easily confounded distinction between measurements and context. Two calibrated polarization monitors provide two spectral observations. A carrier plan states where transmissions are expected, and, under the present receiver assumption, their symbol rates, pulse-shaping parameters and modulation descriptors. A cancellation residual is a derived feature computed from complex co-polarized monitor samples and that plan. These are different information resources. Simply comparing a model with all three resources against a spectrum-only model would not isolate why performance changed. Moreover, a label such as *unauthorized* is defined relative to the declared plan; a measured benefit of plan access on such labels is partly a benefit of supplying the context used to define the task. [E01–E04, E09]

We address three separate questions. **H1** asks whether the canonical dual-polarization monitoring configuration improves on the canonical co-polarization configuration. It adds the cross-polarized spectrum **and its plan mask**, so its interpretation is conditional on plan availability. **H2** asks whether the co-plan-derived residual adds useful representation beyond both observed spectra and masks. It is not a plan-independent cancellation experiment. **H3** asks whether plan masks help when the remaining preprocessing and training choices are matched. Its primary comparison holds a plan-free reference and learning objective fixed and changes only the mask channels, subject to a small input-stem parameter difference. [E09]

Open-set recognition adds a second boundary. A classifier may be correct among the eight trained classes yet confidently assign a nearby untrained waveform to one of them. Conversely, a high unknown score need not mean harmful interference: a legitimate carrier-identification overlay is a separate benign control. Known-interference detection, known-class recognition and unknown rejection must therefore be evaluated with different positive classes and denominators. Open-set and out-of-distribution methods motivate this distinction, but cannot establish the physical meaning of a rejected RF observation (Scheirer et al. 2013; Bendale and Boult 2016; Hendrycks and Gimpel 2017). [E05, E08]

The bounded methodological contribution is an auditable experiment connecting a reference-model dual-polarization payload, receiver-derived features, near-family unknowns and information-matched comparisons. It consists of a versioned scene model; separation of observable inputs from simulator truth; an explicit fifteen-fit minimum design; and a calibration and paired-analysis protocol that preserves ambiguity and adverse outcomes. It is not a claim of first use, a new general-purpose open-set algorithm, a measured satellite benchmark, or demonstrated deployment performance. Existing physical diagnostics motivate the comparisons; the article's recognition Results and outcome-dependent Discussion remain incomplete until the declared experiments are executed. [E01–E10, E16]

2. Related Work

2.1. Satellite interference observations and cancellation

Signal cancellation predates this work as a way to reveal interference hidden beneath a wanted satellite signal (Politis et al. 2017). We use that prior work to motivate residual-based observation, not to claim replication of its receiver or operating point. Our monitor is on the ground after the modeled payload and downlink, and its decision-directed receiver is deliberately limited. Payload distortion and imperfect reconstruction can remain in the residual alongside an interferer. A reduction in wanted-signal power is therefore neither proof that the interferer was preserved nor proof of better classification. The residual must be evaluated through the explicit H2 comparison and its physical controls. [E03, E11]

Henarejos and colleagues study learning with actual Ku-band satellite captures and digitally combined terrestrial signals (Henarejos, Vázquez, and Pérez-Neira 2019). Pellaco and colleagues investigate spectrum prediction and deviation detection with received satellite spectra (Pellaco, Singh, and Jaldén 2019). These studies prevent an overly broad claim that previous satellite learning uses only simulation. They also illustrate why the injection point matters: digitally combining received signals is not the same experiment as propagating an uplink interferer through the wanted nonlinear transponder. We distinguish those observation domains without inferring that one domain universally yields harder or easier recognition. [E02, E18]

On-board CNN interference detection and NGSO-to-GSO detection provide further satellite-specific comparisons

of representation and learning methods (Daoud et al. 2023; Saifaldawla et al. 2024). Their published task definitions and associated source descriptions do not establish a common eight-known-class, dual-polarization, ground-monitor open-set benchmark with the present carrier-plan assumptions. Consequently, their performance numbers are not imported into our tables. A method-level numerical comparison would require compatible observations, labels, splits and lawful data access; no such new comparison is claimed here. [E18]

2.2. Payload models and observation fidelity

ETSI's DVB-S2 reference channel describes the transparent IMUX–amplifier–OMUX structure, while the implementation guidelines provide reference filter data (ETSI 2014, 2021). We use the repository's derived filter responses and digitized nonlinear TWTA curves as traceable reference models. Saleh's classical nonlinear amplifier model is an available analytical alternative, not the default payload substituted into the headline experiment (Saleh 1981). Reference-curve agreement is a consistency target; it is not validation against a newly measured satellite.

The physical coupling model preserves coherent complex mixing between polarization feeds before and after two separate nonlinear paths. Its unitary Jones approximation and effective-linewidth phase-noise model are stated assumptions, rather than full antenna, propagation or oscillator characterizations. Independent instrument calibration corrects the modeled monitor response while retaining propagation leakage. This distinction is necessary to avoid a calibration procedure that inadvertently removes the impairment used to define a class. [E02]

2.3. Open-set learning and RF interpretation

Open-set recognition formalizes the need to reject inputs outside the trained class set (Scheirer et al. 2013). OpenMax is an established deep open-set approach (Bendale and Boult 2016); it is related work, not an implemented baseline in this campaign. Maximum softmax probability is a simple confidence baseline (Hendrycks and Gimpel 2017), energy scores provide a logit-based alternative (Liu et al. 2020), and class-conditional feature distances motivate Mahalanobis scoring (Lee et al. 2018). These scores are compared as fixed post-hoc constructions. We do not reproduce Lee and colleagues' complete multi-layer/input-preprocessing detector, nor train the energy model with auxiliary outliers. [E07, E08]

Residual learning motivates the compact convolutional backbone (He et al. 2016), and center loss motivates training a class-structured embedding (Wen et al. 2016). Our architecture is a small task-specific residual network, not an ImageNet ResNet replica. Our center term uses a batch mean of summed squared distances and optimizes centers with AdamW; its scaling and update rule differ from the original center-loss algorithm. Neither the backbone nor this adapted loss is presented as a novel learning algorithm. The research contribution depends on the physical experiment and fair information contrasts, whose outcomes are still pending. [E07, E09]

Rejection also requires disciplined interpretation. The present calibrated threshold is an empirical known-validation quantile, not a distribution-free guarantee or a conformal predictor. An unknown-family ranking is not a calibrated operating point, and a simulator root-cause label is not proof that the cause is identifiable from the available observation. These distinctions guide the metrics, controls and limitations below. [E01, E08, E10]

3. GEO Bent-Pipe Physical System Model

3.1. Paired multicarrier scenes and transparent payload

Let the two nominal uplink feeds be indexed by $p \in \{c, x\}$. Their complex signals are

$$x_p(t) = \sum_{m \in \mathcal{P}_p} a_{pm} s_{pm}(t) e^{j2\pi f_{pm} t},$$

where $\mathcal{P}_p$ is the declared operator plan and each s_{pm} is a pulse-shaped modulated carrier. QPSK, 8PSK, 16APSK and 32APSK are supported within the repository waveform conventions. This representation does not implement complete DVB-S2 framing, coding, pilot tracking or a standards-compliant modem. The scene generator binds plans, carrier streams, noise streams, link settings, oscillator realizations and missing-carrier state to one scenario group. Class variants share these quantities and alter the registered event. Any infeasible common carrier

placement is handled by a bounded, recorded redraw; unexpected simulator errors propagate rather than silently removing difficult scenes. [E01–E03]

Writing $\mathbf{x} = [x_c, x_x]^T$, a compact expression for the simulated observation is

$$\mathbf{v}(t) = \mathbf{J}_d\,\mathbf{F}\Big[e^{j\phi_s(t)}\mathbf{J}_u\big(\mathbf{x}(t) + \mathbf{i}_u(t) + \mathbf{n}_u(t)\big)\Big] + \mathbf{i}_d(t),$$
$$\mathbf{z}(t) = \widehat{\mathbf{H}}^{-1}\Big[\mathbf{H}e^{j\phi_r(t)}\mathbf{v}(t) + \mathbf{n}_r(t)\Big].$$

The operator $\mathbf{F}$ applies an IMUX, fixed-gain nonlinear amplifier and OMUX separately to each feed. $\mathbf{J}_u$ and $\mathbf{J}_d$ describe physical uplink and downlink coupling; $\mathbf{H}$ is the instrument response and $\widehat{\mathbf{H}}$ its independently estimated calibration. This equation is a block-order description, not a claim that nonlinear multicarrier propagation reduces to a linear matrix channel. The same symbol $\mathbf{n}_r$ includes the receiver/downlink thermal-noise convention used by the implementation. Its per-feed reference is the pre-downlink-mixing OMUX output, and inverse instrument correction transforms that noise as well as the wanted signal. [E02]

The default filters and TWTA are derived reference responses (ETSI 2014, 2021). Input back-off is referenced to the configured saturation convention, not inferred by the classifier. Fixed gain is retained when a planned carrier is absent: active power falls and realized back-off changes. Renormalizing the surviving carriers would remove a physical consequence of the missing-carrier event and would define a different experiment. Overdrive acts on the co-feed payload drive; an uplink interferer enters before coupling and payload processing, whereas the registered downlink ASI event is added after physical downlink mixing and bypasses the wanted payload. [E01, E02, E05]

3.2. Polarization, phase noise and instrument calibration

The symmetric coupling matrix is

$$\mathbf{J} = \begin{bmatrix} c & e^{j\psi}s \\ -e^{-j\psi}s & c \end{bmatrix}, \qquad c^2 + s^2 = 1, \quad c^2/s^2 = 10^{\mathrm{XPD}/10}.$$

This unitary approximation conserves total instantaneous power under ideal basis mixing. XPD describes isolated-source co/cross discrimination; the aggregate ratio of two simultaneously loaded monitored feeds is not an XPD estimate. Uplink and downlink mixing generally do not commute with the nonlinear feed-wise payload. Nominal uplink/downlink XPDs of 30/35 dB, instrument XPD of 25 dB and the 10 dB uplink fault are declared model settings. They are not sampled measurements of a particular satellite fleet or a measured distribution of polarization faults. [E02]

Two independent Wiener phase processes represent a common satellite oscillator and a common receiver oscillator, each shared across both polarizations. For effective linewidth β,

$$\mathrm{Var}[\phi(t + \Delta t) - \phi(t)] = 2\pi\beta\Delta t.$$

The registered linewidths are 100 and 50 Hz. Gap increments connect the sparsely retained snapshots without simulating every skipped high-rate sample. This effective Lorentzian description is not a measured single-sideband phase-noise mask and does not represent rain depolarization, arbitrary frequency-dependent antenna leakage or all receiver synchronization impairments. Static calibration is not oscillator phase tracking. [E02]

Independent full-rank complex pilots fit $\mathbf{Y} = \mathbf{HX}$ by least squares, with rank and conditioning checks before inversion. The baseline uses 64 pilot samples with synthetic complex-noise standard deviation 0.001. Instrument gains and phases are specified in the configuration; only their estimate is used for correction, not the true response matrix. Physical propagation coupling remains in the calibrated scene. These are simulated calibration measurements, not evidence that a real dual-polarization ground station has been calibrated. [E02]

Table 1. Frozen baseline settings. Values are simulation choices unless expressly identified as a reference-model structure.

Quantity	Baseline / range	Interpretation
Channel / simulation / monitor bandwidth-rate	36 MHz / 288 MS/s / 96 MS/s	Finite sampled reference channel
Snapshots / spacing / retained IQ	128 / 10 ms / 4096 samples per snapshot	Sparse schedule, not contiguous 1.28 s IQ
PSD bins / bin spacing	512 / 187.5 kHz	Monitor grid spans 96 MHz
Nominal IBO / overdrive IBO	9–13 dB / 1–4 dB	Fixed-gain payload convention
Uplink and downlink C/N	15–30 dB per path	Declared noise-reference conventions
Uplink / downlink / instrument XPD	30 / 35 / 25 dB	Nominal settings, not measured distributions
Cross-polarization fault	Uplink XPD 10 dB	Controlled impairment
Satellite / receiver linewidth	100 / 50 Hz	Effective Wiener model
Planned-carrier omission	Probability 0.15 per group	One carrier across the two feeds; recorded conditioning

Settings and references: E01–E03. The temporal span describes when short windows are observed; interpreting it as a continuous long record would overstate the available information.

3.3. Registered interference taxonomy

The eight known labels are simulator root causes, with at most one primary event per variant. The independent auxiliary target concerns carrier absence, not an additional primary interference label. Table 2 records which physical intervention defines each known class. This distinction prevents interpreting labels as universal spectral categories. [E01, E03]

Table 2. Known classes and primary interventions. C/I denominators must be retained.

Class	Intervention	Registered level or distinction
Clean	Nominal loaded link	May still have the independent missing-carrier state
CW	Co-feed uplink tone	Total-plan C/I 5–35 dB
Swept CW	Co-feed swept tone	C/I 5–35 dB; sweep span 1 MHz–0.9 channel bandwidth
Unauthorized	Modulated carrier in a plan gap	PSD relative to plan mean: −10 to +3 dB
Overdrive	Co-feed payload drive	IBO 1–4 dB
Cross-pol	Increased uplink coupling	Uplink XPD changed to 10 dB
Uplink ASI	Carrier traversing wanted payload	Geometry-derived PSD C/I; 20–80% plan overlap
Carrier-under-carrier	Carrier inside an active victim band	Total-plan C/I 5–30 dB

The ASI construction uses a simplified single-interferer, common-coverage uplink budget, $C/I_{\mathrm{PSD}} = G_{\max} - G(\varphi) - \Delta\mathrm{EIRP}_{\mathrm{density}}$, with documented reference antenna-pattern branches and explicitly assumed diameter, pointing and EIRP distributions. Its natural level distribution differs from that of the gap carrier. Consequently, high natural-distribution classification accuracy for the ASI/unauthorized pair could reflect level rather than the overlap cue. The dedicated level-matched set is necessary to examine that shortcut; natural and matched results answer different questions. [E01, E06, E12]

The three near-family unknowns are pulsed CW, intermittent CW bursts and downlink ASI. They share mechanisms with trained classes and are deliberately not portrayed as distant arbitrary anomalies. Wideband noise is reported separately because it can be observationally equivalent to an elevated uplink noise floor. The benign `cid_overlay` places carrier-identification signals under transmitted co-feed carriers and has expected decision *clean*; it is excluded from unknown-positive pooling. No test-only family or its outcomes is used as a surrogate unknown for fitting or threshold choice. [E05]

4. Payload-Aware Recognition Methodology

4.1. Spectral inputs and information access

The canonical stored observation contains calibrated spectra $S_c, S_x \in \mathbb{R}^{128\times512}$, plan masks $M_c, M_x \in \mathbb{R}^{512}$ and a co-feed residual spectrum $R_c \in \mathbb{R}^{128\times512}$. Spectra, masks and residuals are stored as float16 and converted to

float32 model inputs during preprocessing. Spectral values are dB of simulated power density, not calibrated field dBm. Masks are broadcast along time and retain their separate relative-level meaning. Time pooling averages linear powers before converting to dB. The frozen pooling factor two yields 64 time positions; spectra are clipped to $[-60, 20]$ dB relative to their reference and scaled by 20. These fixed conventions are not selected from test performance. [E03, E07]

The canonical reference is each scene's observed co-feed mean linear PSD over the co-plan core, defined by mask values above -3 dB, converted to dB. Both polarizations and the residual use this **same** reference. Independent normalization of the feeds would change the relative-level cue H1 examines. This per-scene carrier reference itself uses the plan, so removing only mask input channels would leave a plan-information path in an alleged plan-independent control. [E07, E09]

The canonical views are $X_{\mathrm{co}} = [S_c, M_c]$, $X_{\mathrm{dual}} = [S_c, S_x, M_c, M_x]$ and $X_{\mathrm{residual}} = [S_c, S_x, M_c, M_x, R_c]$, after their declared preprocessing. Simulator truth such as realized IBO, XPD, C/I, source identity, active-carrier masks and event parameters remains metadata for auditing or registered stratification. It is never a learned input. Conditioning plots on such metadata does not turn it into an operationally available detector feature. [E01, E03, E07]

4.2. Observation-derived carrier cancellation

The frozen receiver is P20 `obs-cancel/1`. It consumes calibrated co-feed IQ and declared carrier descriptors: center frequency, symbol rate, roll-off, modulation and the APSK code-rate/ring-ratio identifier. Availability and accuracy of those descriptors are monitoring assumptions. The receiver does not receive transmitted symbols, true activity, event labels or payload settings. It processes every planned carrier, including a carrier that is physically absent. [E03]

For each short monitor block, the receiver resamples to the simulation rate, mixes each planned carrier to baseband and applies a unit-energy SRRC matched filter spanning ten symbols. Oerder–Meyr timing estimation is quantized to the high-rate grid. A 16-point coarse phase search over the modulation symmetry sector initializes hard decisions, followed by three decision-directed complex least-squares gain iterations. The reconstructed carriers are remodulated, summed, decimated and subtracted from the observed co-feed signal:

$$r_c[n] = z_c[n] - \sum_{m \in \mathcal{P}_c} \widehat{z}_{cm}[n].$$

The learned fifth channel is the residual PSD, not the reconstructed symbols or receiver diagnostics. No nonlinear inverse of the TWTA is assumed. [E03]

An edge guard removes 512 monitor samples at each end. The residual therefore uses 3072 samples and eleven Welch segments, versus fifteen segments from 4096 samples for the original spectrum. Identical density scaling and frequency bins do not imply identical estimator variance. Decision errors, filter-induced ISI, cross-polarization leakage, finite boundaries and phase walk remain in the residual. A block-constant complex gain cannot track the frozen phase process within the block; the later theoretical analysis gives phase-noise-only limits around 21.7/22.8 dB for full/guarded windows, not imposed clipping values or measured universal ceilings. [E03, E04]

The separate P21b `B2-features/1` receiver includes fractional timing and a short received-domain FIR and uses a different center-loss convention. It is not silently substituted for P20 and is not a single-variable H2 comparator. Likewise, an oracle event-free twin used in a diagnostic is not part of the operational cancellation input. H2 tests a receiver-derived representation from observed complex IQ and the plan; it does not assert that a deterministic transformation creates information beyond its original IQ inputs. [E03, E04, E11]

4.3. Classifier, objective and fixed training recipe

A full-resolution 5×5 stem feeds four residual stages. Global mean and maximum pooling are concatenated into a $D = 16w$ embedding, followed by an eight-class linear head and, for canonical models, an independent binary auxiliary head. At width $w = 16$, $D = 256$. At width 16, canonical co/dual/residual model parameter counts are 310,505/311,305/311,705, excluding separately learned class centers. H1 and H2 therefore change input-stem capacity as well as the available representation. Retaining the initial frequency grid lets a narrow

spectral line reach a nonlinearity before downsampling; maximum pooling complements the global mean. This is an architectural rationale rather than an observed improvement over another backbone. [E07]

For a training batch of size B, with embedding f_i, class center c_{y_i} and physical missing-carrier indicator a_i, the canonical objective is

$$\mathcal{L} = \mathcal{L}_{\mathrm{CE}} + 0.005\,\frac{1}{B}\sum_i \|f_i - c_{y_i}\|_2^2 + 1.0\,\mathcal{L}_{\mathrm{BCE}}(\widehat{a}_i, a_i).$$

The center sum is not divided by embedding dimension and has no additional one-half factor. Centers share the network's AdamW parameter group and weight decay. This is an explicitly frozen adaptation of center-loss learning (Wen et al. 2016). The auxiliary positive class is a physically absent planned carrier, $a = 1$; it is not general operator-plan/spectrum agreement. Classifier selection uses only primary known-class validation macro-F1, not auxiliary performance. [E07, E10]

The approved recipe uses exactly 30 epochs, batch 32, width 16, time pooling two, peak learning rate 10^{-3}, weight decay 10^{-4}, and the existing cosine OneCycle schedule with cycled momentum. There is no early stopping, clipping, autocast or automatic batch reduction. Both matmul and cuDNN TF32 are explicitly off for FP32 training; this does not promise bit-exact GPU determinism. Train groups alone provide 12,000 known observations and 11,250 optimizer steps per fit. Four-group shards and four-shard blocked-shuffle windows bound memory, with batch remainders carried across windows. Blocked shuffling is not global sample shuffling. [E07, E10]

Canonical training retains shared-channel mirror/time-roll augmentation and K17b plan-mask shift: each training sample is selected with probability 0.15 and shifted continuously within ±0.5 MHz. Both masks move together and normalization is recomputed from the shifted plan, while the stored spectrum and cancellation residual remain unchanged. Validation, calibration and final-test masks are unshifted in the main experiment. This augmentation addresses one belief error; it is not equivalent to recomputing cancellation with a wrong receiver descriptor. [E05, E07, E10]

5. Open-Set Detection and Calibration

5.1. Scores and training-only fitted state

The known prediction is $\widehat{y} = \arg\max_k z_k$ for eight logits z_k. Three unknown scores are reported, always with larger values meaning more unknown-like:

$$s_{\mathrm{MSP}} = 1 - \max_k \mathrm{softmax}(\mathbf{z})_k, \qquad s_E = -\log\sum_k e^{z_k},$$

$$s_M = \min_k (f - \mu_k)^T \widetilde{\Sigma}^{-1} (f - \mu_k).$$

MSP and energy follow established confidence constructions (Hendrycks and Gimpel 2017; Liu et al. 2020). Temperature is fixed at one, without unknown-informed temperature tuning. Energy is sensitive to a common logit offset, unlike MSP, so native score magnitudes are not compared across unrelated models. For Mahalanobis scoring, each μ_k and the pooled within-class covariance are fitted from training embeddings only; no validation or unknown embedding enters their estimation. [E08]

Specifically, Σ uses class-centered residuals and denominator $\max(N - K, 1)$, and

$$\widetilde{\Sigma} = 0.9\Sigma + 0.1\frac{\mathrm{tr}\,\Sigma}{D} I + \delta I, \qquad \delta = \max\left(10^{-6}\frac{\mathrm{tr}\,\Sigma}{D}, 10^{-8}\right).$$

Distances use a Cholesky solve. This single-embedding, shrinkage-regularized score is related to feature-distance detection (Lee et al. 2018), not an exact reproduction of its complete procedure. A positive diagonal floor ensures numerical invertibility even when within-class variability is poorly estimated; it does not establish that a tiny training pilot supplies reliable covariance statistics. [E08]

Mahalanobis is the current **recommendation**, motivated by the center-structured embedding, but is **NOT DESIGNATED** as primary by the researcher. All three scores remain reported regardless of the eventual pre-test designation. Neither favorable pilot ranks nor final held-out outcomes may be used to select the headline scorer retrospectively. [E09, E10]

5.2. Independent rejection and clean-alarm calibration

Each scorer has its own threshold from the known-only `val-calib` role. For ordered calibration scores $u_{(0)} \leq \cdots \leq u_{(n-1)}$, the empirical `higher` 0.95 quantile is

$$\tau = u_{(\lceil 0.95(n-1) \rceil)}.$$

A sample is accepted as known when $s \leq \tau$ and rejected when $s > \tau$. Equality is accepted; ties and finite calibration populations affect realized acceptance. The target 0.95 known acceptance is not a population guarantee, and pooled calibration does not impose 95% acceptance within every known class. Thresholds are specific to the fitted model, score and calibration artifact, and cannot be transported across views merely because their target is identical. [E08, E10]

A separate clean-versus-event alarm uses $s_A = \sum_{k \neq \text{clean}} p_k$, mathematically $1 - p_{\text{clean}}$ but evaluated without subtractive saturation. Its threshold uses **clean** `val-calib` observations only, the same `higher` rule at 0.95, and a strict greater-than alarm. This targets clean false-alarm probability 0.05 and differs from rejection of an untrained family. In particular, known interference is positive for event detection but negative for an unknown-positive ROC calculation. [E08, E10]

5.3. Ranking, operating points and uncertainty

The primary endpoint is near-family pooled unknown-positive FPR@95TPR for the canonical residual view. It is the fraction of known test observations at or above the first grouped-score threshold attaining at least 95% detection of the pooled near-family unknowns. Equal-score observations enter the ranking curve together; this ranking convention differs from strict greater-than rejection at a calibrated operating point. Known negatives include all eight classes, including known interference. This is an evaluation ranking summary whose threshold is determined by the ranked test scores; it is **not** the known-validation calibrated threshold or an operational threshold selected for deployment. Lower FPR@95TPR is better. AUROC, average precision, OSCR and fixed-threshold rejection/acceptance complete the picture. [E05, E08–E10]

For OSCR, the known correct-classification rate counts observations that are both accepted and correctly classified, divided by **all** known test observations. The paired horizontal quantity is the fraction of all unknown test observations accepted as known. Merely accepting a known sample is not correct recognition. AP depends on the pooled prevalence: the complete near-family population has 1,500 unknown observations against 4,000 known observations, whereas a single-family row has 500 unknowns against the same 4,000 knowns. The wideband-noise and benign-CID controls are separate populations. Closed-set macro-F1, per-class recall and confusion are computed without pretending that a binary event detector predicts eight classes. [E05, E08]

Intervals resample whole scenario groups, preserving their paired class variants and unknown families. The same bootstrap resample is shared across compared arms and seeds. These percentile intervals condition on the frozen weights, feature statistics and thresholds; they do not include training, calibration, simulator or field-distribution uncertainty. Seeds 0, 1 and 2 are reported separately with a descriptive mean and spread, not a three-seed confidence claim. The accepted design does not support a confirmatory superiority claim or unadjusted multiplicity inference from many class/bin/scorer comparisons. Empty or undefined metrics remain unavailable with counts and reasons; they are never encoded as zero success. [E09, E10]

6. Experimental Design and Ablations

6.1. Scenario roles, pairing and exposure

Table 3. Full intended populations. Counts describe the frozen generator, not completed production.

Role	Groups	First group ID	Observations / use
Train	1500	3000000	12,000 known; weights and feature statistics
Validation-selection	150	3100000	1,200 known; checkpoint selection only
Validation-calibration	150	3150000	1,200 known; rejection and clean alarm
Natural test	500	3200000	4,000 known + 2,500 test-only/control variants
Level-matched test	150	3300000	450 clean/ASI/unauthorized variants

The total is 21,350 observations: 18,400 known variants across the four known roles, 2,500 test-only/control variants and 450 matched variants. Registration here means versioned repository commitments and dated researcher decisions, not registration in an independent external registry. Every group remains in one role. Sharing carrier/noise and event-control realizations across a group's variants is deliberate pairing, not independent replication. The group, rather than each of its eight images, is the resampling unit. Unknown families are excluded from training, checkpoint selection, feature-statistic fitting and calibration. [E01, E03, E05, E06, E10]

The preserved test population is **not pristine**. Groups 3200000/3200001 and matched groups 3300000/3300001 were exposed in earlier pilots; group 3200000 also has a committed observation-derived display. Every final report must therefore retain the 500/150-group headline and show 498/148-group sensitivity rows excluding those IDs, without choosing exclusions by outcomes. This disclosure cannot prove that earlier exposure had no influence on design decisions. It instead makes that limitation auditable and prevents a false claim that freeze-before-read erases historical access. This manuscript reads only already committed summaries and figures. [E10]

6.2. H1, H2 and H3 as distinct estimands

Table 4. Registered contrasts. The sign is richer minus comparator; negative FPR@95TPR differences favor the richer arm.

Question	Required contrast	Held fixed / changed	Interpretation limit
H1	Dual − co	Same canonical recipe; adds cross PSD and cross mask	Dual monitoring with plans, not isolated spectrum-only information
H2	Dual-residual − dual	Same canonical recipe; adds co-plan-derived residual	Plan-conditional receiver representation
H3	Dual-planmask − dual-noplan	Same plan-free reference/objective; adds two masks	Mask-access benefit, with a small stem-capacity difference

For H3, both arms use one fixed reference: the training mean of the per-observation 90th frequency percentile of time-averaged co-feed PSD **in dB**. They share that plan-independent scalar at validation and test time. Neither arm uses the auxiliary plan-consistency objective or plan-shift augmentation, and neither uses a residual. The no-plan path drops mask/residual arrays before preprocessing and rejects plan-derived settings. This closes information paths that would remain if one merely hid the mask channels of a canonically normalized model. The control is global-reference plan-independent, not a stronger per-scene plan-free normalization variant that was never added to the required roster. [E07, E09]

The two matched arms share seeds, data order, augmentation streams, optimizer/schedule, center-loss convention and all non-stem tensor initialization. The different input dimensionality necessarily changes the first convolution; at width 16 the dual mask arm has 800 additional stem parameters. Equal nominal width is therefore not exact parameter equality. The initialization procedure re-seeds non-stem modules independently of stem shape, preventing an avoidable RNG-order confound. Relative to the canonical recipe, both matched arms change exactly three declared settings: fixed plan-free rather than per-scene carrier reference, no auxiliary plan head, and no plan shift. Consequently, canonical dual minus dual-planmask is only a descriptive decomposition, and canonical dual minus dual-noplan does not isolate mask value. [E09]

For a metric m, the paired seed-specific effect is $\Delta_{h,s} = m(\text{richer}_{h,s}) - m(\text{comparator}_{h,s})$. Both point estimates and bootstrap differences use aligned populations and the frozen scorer designation. Selection of a favorable seed, scorer, subgroup or optional arm after viewing protected results is excluded. A5 plan corruption tests sensitivity of an already trained plan-using system; it does not replace H3's learned, matched plan-access comparison. Likewise, H1 without plans would require the optional co-noplan fits, which are not part of the fifteen-fit minimum. [E09, E10]

6.3. Near-family and level-matched controls

Downlink ASI reuses the known uplink ASI group's event draw so that geometry, modulation, rate, roll-off, placement and requested PSD C/I are paired. Injection before versus after the wanted payload is the intended distinction. However, the downlink event is scaled by realized wanted-feed power while uplink interference uses nominal unit loading. A missing co-feed carrier therefore creates a registered level drift; this is not an ideal all-conditions level-matched causal intervention. The accepted generator-derived metadata identify 35 co-feed missing groups and their 465-group complement; 43 cross-feed drops make 78 drops in total. The 500/35/465 report is distinct from the 500/498 exposure sensitivity. The later production metadata comparison is mandatory and **NOT RUN**. [E05, E10]

The separate ASI/unauthorized set uses 150 groups with shared requested $C/I_{\text{PSD}} \sim U(4, 13)$ dB, symbol rate, roll-off, modulation and link conditions, alongside a clean twin. Its primary bin is 7–10 dB, with 4–7 and 10–13 dB neighboring bins. Frequency placement remains different by class definition: overlap versus free gap. Equal requested injected power does not force equal observed PSD rise, because the clean background and nonlinear interactions differ between the two bands. Results must retain full pair-class denominators, including clean/other-known predictions and rejected samples; conditioning only on predictions of the two desired classes could conceal failure. [E06]

Joint placement feasibility also conditions the matched population: its carrier rates are at most 6 Mbaud, whereas natural test-role ASI spans 1–28.8 Mbaud, and the matched 10–13 dB bin extends beyond the natural unauthorized level range. An ASI variant can overlap a planned but physically absent carrier, so its relation to actual transmitted occupancy must be reported separately. P22's metadata-only pre-draw found this intersection in 5 of 150 groups; that count is not a completed production observation measurement. These limitations prevent equating natural-versus-matched changes with a pure level intervention, even when the two arms within a matched group share requested power. [E06]

Additional registered interpretation controls include CuC level bins 5–15, 15–25 and 25–30 dB; noise/back-off stratification; benign CID clean acceptance and false CuC calls; and overdrive scenes whose out-of-plan regrowth remains below the clean-calibrated threshold. These are mechanism-specific views of the same frozen predictions, not opportunities to change labels after observing difficulty. Source truth may define reporting strata but is not an extra model input. Receiver-consistent plan-error experiments, full leave-one-known-out sweeps, alternative cyclic features and linear-to-nonlinear transfer remain outside the minimum campaign; none has a newly executed result here. [E01, E06, E10, E14]

6.4. Required fits and pre-test freeze

The minimum roster is nine canonical fits, $\{\text{co}, \text{dual}, \text{dual_residual}\} \times \{0, 1, 2\}$, and six matched dual fits, $\{\text{dual_planmask}, \text{dual_noplan}\} \times \{0, 1, 2\}$. Optional matched co-planmask/co-noplan fits would add six and are deferred. They must not be chosen after the first protected result. Thirty epochs are run completely; the first highest primary validation macro-F1 selects the checkpoint. An interrupted production fit is incomplete and restarts from epoch zero in a fresh attempt rather than splicing optimizer, scheduler or RNG state. [E09, E10, E16]

Before any separately authorized final observation read, the campaign binds all fifteen checkpoint, scorer and calibrator identities, selected epochs, configuration, population/role identity, source and effective runtime, the researcher's primary-scorer designation, and the fixed canonical-then-H3-dual evaluation sequence. The optional co controls are recorded as deferred. A scorer designation is not itself permission to read data. Current code provides the campaign admission/custody interface, but the separately authorized production held-out inference integration remains absent. The present manuscript neither implements that reader nor opens the protected set. [E16]

7. Existing Simulated Evidence and Results Specifications

7.1. What has actually been measured

The observations in Table 5 are already committed **simulation diagnostics**, not new experiments and not measured satellite data. They motivate questions about available cues without answering H1–H3. The older single-polarization runs omit the later dual-polarization and phase-noise blocks; P20 cancellation uses a different population. Their numerical values are not combined into a common detector comparison. Exact aggregate pointers and figure hashes are supplied in E11–E13. [E11–E13]

Table 5. Existing physical evidence, with its observation boundary.

Diagnostic	Existing population	Recorded quantity	Permitted interpretation
Natural uplink ASI	120 single-pol groups; IBO 11 dB; C/N 25 dB; no phase noise	Median PSD C/I 20.3 dB	Result of the assumed geometry distribution
CuC at total-plan C/I 20 / 30 dB	24 paired single-pol groups at each level; same fixed link	Median twin PSD contrast 0.373 / 0.041 dB	Weak spectral cue under this model; no Pd estimate
P20 cancellation	64 observations, 8 groups across four roles	Class mean active-core depth 12.78–16.47 dB	Receiver behavior; not full-band depth or recognition
P20 CuC twin contrast	Same 8 groups	Median 0.986 to 6.461 dB before/after; 8/8 increase	Offline truth-selected victim-band diagnostic

The event/no-event contrast is $10\log_{10}(\overline{P}_{\text{event}}/\overline{P}_{\text{twin}})$ in a diagnostic region. The twin and truth-selected victim band are unavailable to a live monitor. The ASI PSD C/I and CuC total-plan C/I denominators are different and must not share an unlabeled horizontal axis. At high C/I, a small twin contrast motivates testing alternative representations, but does not prove an information-theoretic impossibility or a neural detection probability. [E11, E12]

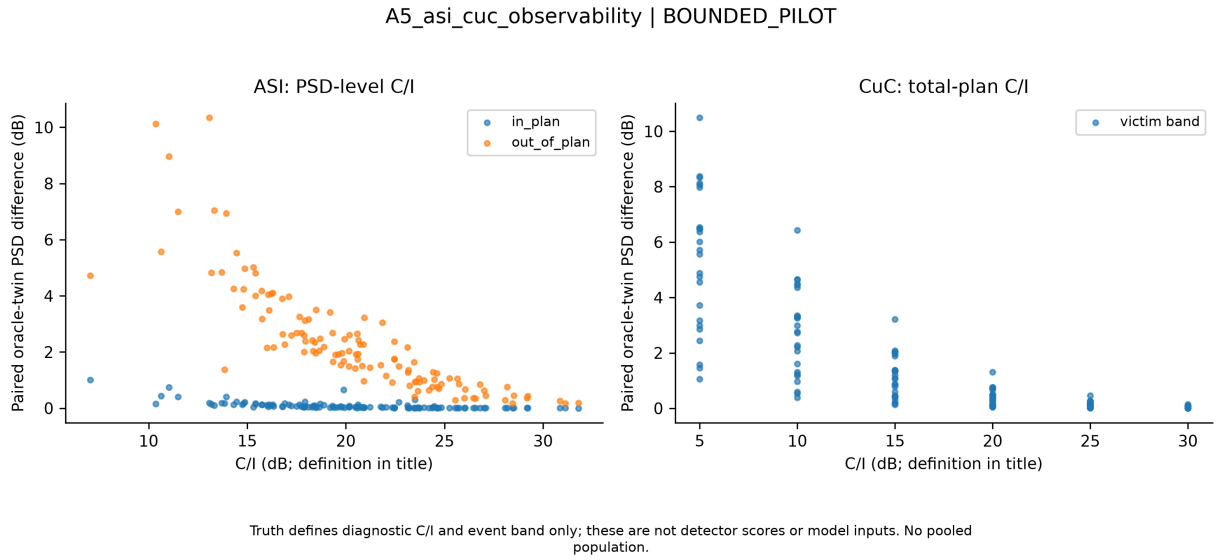


Figure 1: Reused physical visibility diagnostics, not recognition results. ASI uses 120 single-polarization groups and PSD-referenced C/I; CuC uses 24 groups at each level and total-plan C/I. Both use the older fixed IBO 11 dB, uplink/downlink C/N 25 dB configuration without phase noise. Contrasts use paired event-free twins and diagnostic bands that are not operational model inputs. The panels have different interference references and are not a common detector-performance curve. Reused byte-for-byte from the validated figure package; E12–E13.

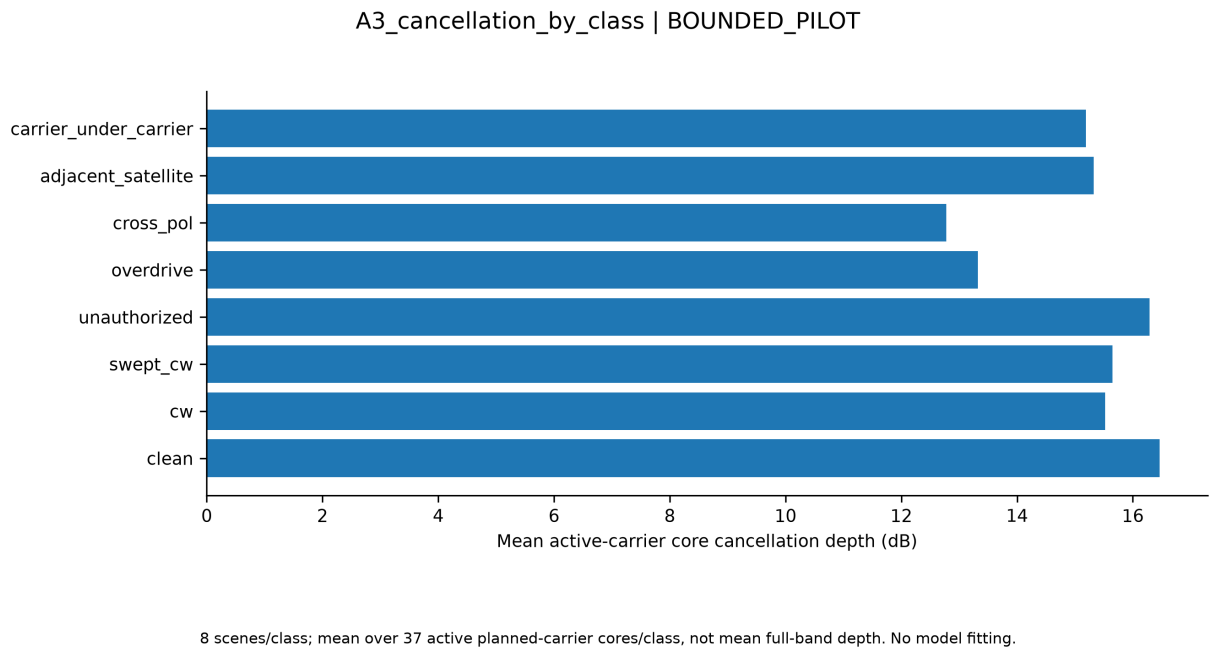


Figure 2: Existing P20 class-wise cancellation diagnostic: 64 observations in eight paired groups across four roles, eight observations per class and 37 active planned-carrier cores per class. Mean occupied-core PSD reduction is not full-band cancellation depth, Pd or recognition improvement. The historical diagnostic included test-role pilot groups; that exposure is disclosed, and this draft reuses only the committed aggregate figure. The bars are class means; no confidence interval is displayed. Reused byte-for-byte; E11, E13.

Historical small neural pilots include chance-level/degenerate predictions and uninformative auxiliary accuracy where no positive missing-carrier cases occurred. Bounded train/calibrate/freeze and C2 invariance exercises verify data flow and control isolation, not relative recognition quality. They are not substituted for any entry in the following production tables. In particular, neither a favorable physical residual contrast nor a successful checkpoint reload supplies an observed H2 effect. [E09, E11, E15]

7.2. Production recognition results — NOT RUN

Table 6. Required fit inventory and publication table skeleton. Every measurement cell is explicitly NOT RUN. All three scorers must be retained; "primary" remains NOT DESIGNATED.

View	Seeds	Known accuracy / macro-F1	Near FPR95: MSP	Near FPR95: energy	Near FPR95: Mahalanobis
Co	0, 1, 2 separately	NOT RUN	NOT RUN	NOT RUN	NOT RUN
Dual	0, 1, 2 separately	NOT RUN	NOT RUN	NOT RUN	NOT RUN
Dual-residual	0, 1, 2 separately	NOT RUN	NOT RUN	NOT RUN	NOT RUN
Dual-planmask	0, 1, 2 separately	NOT RUN	NOT RUN	NOT RUN	NOT RUN
Dual-noplan	0, 1, 2 separately	NOT RUN	NOT RUN	NOT RUN	NOT RUN

The final table expands every seed into its own row and adds a separately labeled descriptive mean/spread. Companion tables must provide AUROC/AP/OSCR, known acceptance, accepted-known accuracy, unknown rejection and clean false alarms with numerators and denominators. Exposure-excluded and K17d rows retain separate labels. The linked display specification defines populations, axes and aggregation without inventing measurements. [E09, E10, E17]

Table 7. Paired hypothesis-effect skeleton. Units are percentage points for FPR95; intervals condition on fixed fitted state.

Contrast	Seed 0	Seed 1	Seed 2	Descriptive seed mean	Paired group-bootstrap interval
H1: dual − co	NOT RUN	NOT RUN	NOT RUN	NOT RUN	NOT RUN
H2: dual-residual − dual	NOT RUN	NOT RUN	NOT RUN	NOT RUN	NOT RUN
H3: dual-planmask − dual-noplan	NOT RUN	NOT RUN	NOT RUN	NOT RUN	NOT RUN

These are publication specifications, not empty evidence for a favorable direction. A positive, negative, null-looking or uncertain effect must be retained as observed. Ranking metrics and clean-calibrated operating metrics can disagree and require separate interpretation. Outcome-dependent Discussion is deliberately unwritten: no claim about which information source helps is justified until Table 7 and the corresponding operating-point/ambiguity controls are populated. [E09, E17]

7.3. Minimum experiments to complete Results and Discussion

First, after separate execution authorization under the frozen protocols, generate and verify the known, test-only/control and level-matched artifacts, retaining the distinct populations and metadata-only D8 cross-check. Second, execute the fifteen declared 30-epoch fits, select on validation-selection, fit feature statistics from train only, calibrate on known validation-calibration and freeze every fitted object. Third, fix the primary scorer and evaluation order before the first separately authorized protected read, then evaluate canonical and H3-dual inventories once using the same natural and matched populations. Fourth, compute the offline paired contrasts and the required exposure, K17d, class/level and benign-CID reports, and populate the defined tables/figures with counts and conditional uncertainty. No additional fit is required for these reporting strata. [E09, E10, E16, E17]

These are the minimum experiments for the **narrow H1–H3 manuscript**, not completion of every historical B2 extension. Full LOKO, SCF, amplifier-transfer/C3 and receiver-level plan-error studies must remain labeled unexecuted unless separately specified and authorized. The optional co-only H3 controls are not silently added to the headline. If future authors claim superiority to CFAR, EDSC or expert-feature classifiers, those comparators must also be run on the identical frozen production populations; bounded historical baseline figures cannot supply that comparison. The current draft makes no such superiority claim. [E01, E09, E14, E17]

8. Threats to Validity and Reproducibility

8.1. Physical and construct validity

The simulator implements a controlled reference channel, not a calibrated digital twin of a named GEO payload. Digitization and finite FIR responses limit the filter/amplifier representation. The Jones approximation excludes general lossy or frequency-dependent coupling; effective Wiener linewidths are not measured oscillator spectra. Assumed geometry, pointing and loading distributions do not establish operational prevalence. The experiment addresses conditional behavior within those assumptions, and field transfer requires independent measured evidence. Results from measured GNSS or solid-state PA tasks are not inserted as validation of this GEO recognition system. [E02, E12, E18]

Root-cause labels can exceed what is identifiable in the selected observable. Weak CuC, noise-floor changes and nominal-looking overdrive are deliberate hard cases. ASI/unauthorized placement is defined relative to a plan, while a downlink event can resemble an uplink event despite a different causal path. Better classification under the simulator may exploit correlates of its sampling rules. It does not establish unique source attribution, authorization judgment without an operator plan, or identification of an interfering satellite. The level-matched and missing-carrier controls expose specific confounds; they do not exhaust all possible shortcuts. [E01, E05, E06, E12]

8.2. Receiver and information-assumption validity

Cancellation assumes accurate carrier descriptors and shares waveform/constellation conventions with the simulator. P20 records unresolved APSK start-angle standard verification and grid-aligned timing conditions; self-consistency should not be presented as independent modem validation. A stronger real receiver might track phase and equalize better; a real monitor with stale descriptors might perform worse. Neither direction is established here. H2 measures this particular representation and receiver. H1 also changes plan-mask access, and H3 retains a small stem-capacity difference and a global, rather than per-scene, plan-free reference. These delimit the estimated effects even if the resulting tables show a large gap. [E02–E04, E09]

Plan-mask augmentation is especially easy to overinterpret. Its stored residual is fixed, whereas an erroneous receiver plan can change cancellation itself. Historical A5 diagnostics distinguish mask-only and receiver-consistent interventions, but their tiny chance-level models do not establish robust recognition. A plan sensitivity claim requires its own receiver-consistent experiment and cannot be inferred from H3 or from a no-change argmax in an execution pilot. [E04, E09, E14]

8.3. Statistical validity and prior exposure

Paired variants are dependent by design, calibration is finite and three training seeds are a small sample. Group bootstrap intervals condition on already selected/fitted objects and omit several sources of uncertainty. Numerous scorers, families, bins and contrasts increase interpretive flexibility. All fixed scores and adverse outcomes must be reported; seed averages are descriptive and no post-hoc winner is promoted. The 95% known-acceptance target and 5% clean-alarm target do not guarantee test behavior, while FPR@95TPR summarizes a different thresholding question. [E08–E10]

Historical test exposure is a material limitation, not a footnote cured by a new artifact hash. Both complete and identity-excluded sensitivity populations are required, and neither replaces the separately labeled co-feed-missing analysis. A sign-consistent result across these rows would improve interpretability but would not retroactively create an untouched external test. Subsequent design changes motivated by final outcomes require new, prospectively separated evidence rather than rebranding the current test population. [E10]

8.4. Reproducibility, artifacts and access

Generator requests bind source, configuration, role populations, calibration and checksummed shards. Training and calibration markers bind data, source, runtime and fitted-state identities; stale checkpoints are rejected instead of relabeled. Effective Torch/CUDA/cuDNN/device/driver and TF32 settings must be recorded for production execution. An interrupted fit is incomplete, and no missing or unavailable measurement is interpreted as zero. Such provenance makes a result auditable; it cannot establish that the assumed physical model is true or that a storage device survived an untested failure. [E03, E07, E10, E16]

This manuscript package contains Markdown, bibliography, traceability records, document-only validation/build tools and reused aggregate figures. It requires no raw IQ, protected dataset, checkpoint, simulator run or fitting operation to build. The existing comparative-methods manuscript and independent literature review remain separate and unchanged. Public primary references establish cited method identities and bounded prior-work statements, while the source audit records full-text versus metadata-only access. Authors, affiliations, venue suitability and publication approval remain human decisions. No claim of systematic literature completeness or priority is made. [E13, E16, E18]

9. Conclusion

The scientific question is how explicitly stated observation resources change a fixed open-set GEO recognition task, rather than whether a single architecture can label every possible interference cause. The proposed experiment separates dual-monitor access, a plan-conditioned receiver residual and mask access under matched preprocessing. Its role isolation, known-only calibration, near-family grouping and paired analyses provide a concrete basis for answering those questions. Existing physical diagnostics explain why the questions are nontrivial; they do not answer them. Production-scale recognition results, the primary-scorer designation and final protected evaluation remain pending. The manuscript is therefore a substantive methods-and-design draft with honest

evidence boundaries, ready for completion by the specified experiments rather than by invented performance claims. [E01–E18]

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